

FitMind AI Gym OS - Complete Implementation Guide

What You’re Building

A complete AI-powered gym management platform that replaces 5+ tools with one intelligent system. Gyms become dependent on your platform because every operational touchpoint flows through it.

Your Deliverables (3 Products in 1)

1. Web Dashboard (Next.js 15)

For: Gym owners, managers, coaches **What it does:** Complete command center for the entire gym operation **Why they can’t leave:** All their data, schedules, billing, and AI insights live here

2. Member Mobile App (React Native)

For: Gym members **What it does:** Book classes, check in, track progress, pay bills, refer friends **Why they can’t leave:** Their workout history, streaks, and AI coach live here

3. AI Backend (Python/FastAPI)

For: Runs 24/7 behind the scenes **What it does:** Predicts churn, optimizes schedules, recovers payments, nurtures leads **Why they can’t leave:** The intelligence that makes everything else work

Tech Stack (Production-Ready)

Layer	Technology	Why
Frontend	Next.js 15 + Tailwind + shadcn/ui	Server Components, fastest framework, beautiful UI
Mobile	React Native + Expo	One codebase, iOS + Android, branded apps
Backend	Python + FastAPI	AI-native, async, auto-generated docs
Database	PostgreSQL 15	ACID, JSONB for flexibility, battle-tested
Cache	Redis	Real-time data, sessions, queues
AI/ML	OpenAI + scikit-learn	NLP for chat, ML for predictions
Payments	Stripe	Industry standard, subscriptions, retries
Messaging	Twilio (SMS) + SendGrid (Email)	Reliable delivery
Hosting	Vercel (Frontend) + Railway/Render (Backend)	Deploy in minutes

Project Structure

```
fitmind-ai/
├─ apps/
│   └─ web/                                # Next.js 15 - Owner Dashboard
│       └─ app/
│           └─ (dashboard)/
│               └─ page.tsx                # Main dashboard
│               └─ members/
│               └─ classes/
│               └─ billing/
│               └─ retention/
│               └─ ai-assistant/
│           └─ api/
│           └─ login/
│       └─ components/
│       └─ lib/
│       └─ package.json
│
├─ mobile/                                # React Native - Member App
│   └─ src/
│       └─ screens/
│           └─ HomeScreen.tsx
│           └─ BookingScreen.tsx
│           └─ ProfileScreen.tsx
│           └─ CheckInScreen.tsx
│       └─ components/
│       └─ navigation/
│       └─ api/
│   └─ package.json
│
├─ packages/
│   └─ api/                                # Shared API client (generated from OpenAPI)
│   └─ ui/                                 # Shared UI components
│   └─ types/                              # Shared TypeScript types
│
├─ services/
│   └─ ai-engine/                          # Python FastAPI + AI models
│       └─ app/
│           └─ main.py
│           └─ routers/
│               └─ retention.py
│               └─ scheduling.py
│               └─ billing.py
│               └─ assistant.py
│           └─ models/
│               └─ churn_predictor.py
│               └─ demand_forecaster.py
│               └─ pricing_optimizer.py
│           └─ services/
│               └─ stripe_service.py
```

```

| | | | | └─ twilio_service.py
| | | | | └─ sendgrid_service.py
| | | | | └─ openai_service.py
| | | | └─ workers/
| | | |   └─ retention_worker.py
| | | |   └─ billing_worker.py
| | | |   └─ scheduling_worker.py
| | └─ requirements.txt
| └─ Dockerfile
|
└─ database/
    └─ migrations/
    └─ seeds/
    └─ schema.sql
└─ docker-compose.yml
└─ turbo.json
└─ README.md

```

Build Order (16-Week MVP)

Phase 1: Foundation (Weeks 1-4)

Goal: Get a gym owner to log in and see their dashboard

```

# Week 1: Database + Auth
- Set up PostgreSQL with schema
- Create Next.js project with auth
- Build login/register pages
- Connect to database

# Week 2: Members + Classes
- Member CRUD (list, create, edit, delete)
- Class types management
- Basic class scheduling

# Week 3: Bookings + Check-In
- Member booking flow
- Waitlist management
- QR code check-in

# Week 4: Dashboard Analytics
- Revenue KPIs
- Attendance charts
- Member growth

```

Phase 2: Money (Weeks 5-8)

Goal: Process real payments and protect revenue

```
# Week 5: Stripe Integration
- Recurring subscriptions
- Invoice generation
- Payment history

# Week 6: Failed Payment Recovery
- Auto-retry logic
- Dunning emails
- Payment method updates

# Week 7: Billing Dashboard
- MRR tracking
- Outstanding invoices
- Revenue reports

# Week 8: Member App v1
- React Native setup
- Login + profile
- Class booking
- Check-in
```

Phase 3: AI Magic (Weeks 9-12)

Goal: The “wow” factor that makes them addicted

```
# Week 9: Churn Prediction
- Train model on member behavior
- Risk scoring algorithm
- Alert generation

# Week 10: Automated Retention
- Triggered messages (SMS/Email/Push)
- Win-back campaigns
- Milestone celebrations

# Week 11: AI Assistant
- OpenAI integration
- Natural language commands
- Auto-scheduling

# Week 12: Smart Recommendations
- Class recommendations for members
- Pricing optimization
- Demand forecasting
```

Phase 4: Growth (Weeks 13-16)

Goal: Turn members into marketers

```
# Week 13: Lead Capture
- Website forms
```

- Walk-in digital waivers
- Social media integration

Week 14: Lead Nurturing

- Automated email sequences
- AI lead scoring
- Tour scheduling

Week 15: Referral Engine

- Member referral codes
- Reward tracking
- Social sharing

Week 16: Polish + Launch

- White-label branding
- Multi-location support
- Performance optimization
- Documentation

The “Dependency Loop” (Why They Can’t Cancel)

Every feature is designed to trap data and create switching costs:

MEMBER SIGNS UP

- Digital waiver stored (legal requirement, can't lose)
- Payment method saved (Stripe token, hard to migrate)
- App downloaded (push tokens, habit formed)

MEMBER BOOKS CLASS

- Schedule optimized by AI (lose the intelligence)
- Waitlist position tracked (member expectations set)
- Reminder sent (member depends on notifications)

MEMBER CHECKS IN

- Visit history built (can't recreate)
- Streak tracked (psychological investment)
- Milestones celebrated (emotional attachment)

MEMBER MISSES DAYS

- AI detects risk (proprietary model)
- Personalized message sent (can't replicate tone)
- Offer generated (revenue protected)

MEMBER REFERS FRIEND

- Referral code tracked (reward system)
- Social proof built (network effects)
- Credits accumulated (financial stake)

Average trapped data value after 6 months: \$50,000+ Cost to migrate to another system:

3-6 months of manual work

Pricing Strategy

Tier	Monthly	Annual (Save 20%)	Members	Features
Starter	\$99	\$950	Up to 200	Core scheduling, billing, basic AI
Growth	\$249	\$2,390	Unlimited	Full AI retention, branded app, analytics
Enterprise	\$499	\$4,790	Multi-location	White-label, API, custom AI

Revenue per gym scales with their success. A 500-member gym on Growth = \$249/mo. At 1,000 members, they upgrade to Enterprise = \$499/mo.

Getting Your First 10 Customers

Step 1: Build the Demo (Week 1)

- Use the interactive widget above as your pitch deck
- Record a 3-minute Loom video walking through it
- Create a simple landing page: “AI That Runs Your Gym”

Step 2: Find Beta Gyms (Week 2)

- Post in Facebook groups: “Gym Owners”
- DM 50 gym owners on Instagram
- Attend 1 local fitness expo
- Offer: “3 months free, then \$99/mo”

Step 3: Launch with 1 Gym (Week 3-4)

- Onboard them manually
- Fix bugs in real-time
- Document every pain point
- Get a video testimonial

Step 4: Scale to 10 (Months 2-3)

- Use testimonial in outreach
- Offer referral bonuses (\$100/gym referred)
- Build case studies with numbers
- Launch on Product Hunt

Step 5: Hit \$10K MRR (Month 6)

- 40 gyms × \$249 = \$9,960/mo
- Add Enterprise tier for chains
- Hire first support person
- Start thinking about Series A

Key Metrics to Track

Metric	Target	Why It Matters
Activation Rate	>80%	Gym owner completes setup within 7 days
Daily Active Users	>60%	Members open app at least 3x/week
Churn Reduction	-30%	Your AI saves them members
Payment Recovery	>85%	Failed payments successfully retried
NPS Score	>50	Gym owners love the product
Net Revenue Retention	>120%	Gyms expand/upgrade over time

Next Steps (Start Today)

1. Clone the repo structure (or use the files provided)
2. Set up PostgreSQL locally
3. Run the schema (psql -f fitmind_database_schema.sql)
4. Create Next.js app (npx create-next-app@latest apps/web)
5. Build the login page (simple email/password)
6. Connect to database (Prisma ORM recommended)
7. Show it to 3 gym owners (get feedback, iterate)

Files Included

File	What It Is	Use It For
fitmind_database_schema.sql	Complete PostgreSQL schema	Run this to create your database
fitmind_api_spec.yaml	OpenAPI 3.0 specification	Generate API clients, documentation
gym_retention_crisis_dashboard	Interactive visualization	Show gym owners why they need AI
gym_ai_impact_dashboard	Before/after metrics	Prove ROI in sales conversations
Interactive Widget	Live prototype	Demo to prospects, get feedback

Support & Community

- **GitHub:** github.com/fitmind-ai/gym-os
- **Discord:** discord.gg/fitmind
- **Email:** founders@fitmind.ai

Built with AI. Powered by data. Designed to make gym owners unstoppable.